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SUB. : FUNDAMENTALS IN AI ML

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TOPIC : **AI Loan Approval System**

PROJECT REPORT

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1. Introduction

In today's world, banks receive a huge number of loan applications every day. Checking each application manually can be slow and sometimes unfair due to human bias.

To solve this problem, I developed an **AI Loan Approval System** using Machine Learning. This system helps predict whether a loan should be approved or not based on important applicant details like income, credit score, and employment status.

The main idea behind this project is to make the loan approval process **faster, smarter, and more reliable**.

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2. Objectives

The main goals of this project are:

- To build a model that can predict loan approval
- To understand which factors affect loan approval the most
- To visualize data for better understanding
- To allow users to enter their own details and get instant predictions

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🔹 3. Technologies Used

This project is built using:

- **Python** (main programming language)

Libraries used:

- **NumPy** – for numerical calculations
- **Pandas** – for handling data
- **Matplotlib** – for creating graphs
- **Scikit-learn** – for building the machine learning model

🔹 4. Dataset Description

For this project, I created a small dataset manually to simulate real-world loan data.

It includes the following features:

- **Income** – how much the applicant earns
- **Credit Score** – indicates financial reliability
- **Age** – age of the applicant
- **Loan Amount** – how much loan is requested
- **Employment** – whether the person is employed (1 = Yes, 0 = No)
- **Approved** – final decision (1 = Approved, 0 = Rejected)

🔹 5. Methodology

5.1 Data Collection

Instead of using an external dataset, I created a sample dataset using Python.

5.2 Data Preprocessing

I added a new feature:

- **Income-to-Loan Ratio = Income / Loan Amount**

This helps the model understand whether the applicant can realistically repay the loan.

5.3 Model Selection

I used the **Random Forest Classifier** because:

- It works well with different types of data
- It reduces overfitting
- It gives good accuracy
- It shows feature importance

5.4 Model Training

- The dataset is divided into:
  - 80% training data
  - 20% testing data
- The model learns patterns from the training data

5.5 Model Evaluation

To check performance, I used:

- **Accuracy Score**
- **Confusion Matrix**

🔹 6. <u>Data Visualization</u>

To better understand the data, I created some graphs:

- **Income vs Loan Approval**
  → Shows how income affects approval
- **Credit Score vs Loan Approval**
  → Higher scores usually mean higher chances of approval
- **Feature Importance Graph**
  → Shows which features are most important in decision-making

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🔹 7. Results

The model performs well on the dataset and is able to predict loan approval correctly in most cases.

Key observations:

- Applicants with **higher income** are more likely to get approval
- A **good credit score** plays a very important role
- Being **employed** increases the chances of approval

🔹 8. User Input Prediction

One of the best parts of this project is that users can enter their own details such as:

- Income
- Credit Score
- Age
- Loan Amount
- Employment Status

Based on this, the system instantly predicts:

- ✅ Loan Approved
- ❌ Loan Rejected

🔹 9. Advantages

- Saves time compared to manual checking
- Reduces chances of human bias
- Easy to use and understand
- Can be expanded for real-world use

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10. Limitations

- The dataset is small and not real
- Real banking systems are more complex
- Limited number of features used

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11. Future Enhancements

In the future, this project can be improved by:

- Using real-world datasets
- Adding more features like education, loan history, etc.
- Converting it into a web or mobile application
- Using advanced models like Deep Learning

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12. Conclusion

This project shows how Machine Learning can be used in real-life financial decision-making.

The AI Loan Approval System successfully predicts loan approvals and helps in understanding the key factors behind decisions. It is a simple but powerful example of how AI can make processes faster and smarter.

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13. References

- Scikit-learn Documentation
- Python Official Documentation
- Online Machine Learning Tutorials